

Ethical and Legal

Privacy and Security

In terms of data collection and storage within our website we aim to comply fully with GDPR rules and regulations. We have already made steps to meet GDPR by complying with the data minimization principle (Article 5(1)(c) of the GDPR), i.e., we are only collecting the minimal relevant data from the user that we need for the website to function correctly. This data being the user's username, email, and password. We aim to further comply with GDPR in the future by securely storing this data to meet with the integrity and confidentiality principle (Article 5(1)(f) of the GDPR), for example, storing hashed passwords instead of plaintext passwords.

User accounts are protected by the user's password which prevents trivial attempts to gain unauthorized access to another user's account. This protects the user's personal data and means that untrusted users cannot log into a trusted user's account and perform actions that only trusted users should be able to perform.

Accessibility

We have tried to make our website accessible to a wide variety of users through a simple and intuitive frontend design. For most of our web pages we tried to keep the design as simple as possible, only adding the bare minimum, so that users don't get overwhelmed by a multitude of buttons, images, or input fields. For the few buttons and input fields we do have, we have given them clear labels and have made them big enough to be easily seen. This helps give the user a better understanding of the purpose of each page and how they should use them. Another way that we have tried to make the website accessible is through the use of colours. Elements on the website have a contrasting colour to the background, and certain elements have borders, so that they can be easily seen and differentiated from the background. Certain usable elements, like buttons, also change colour when hovered over so that users can clearly see which button they are about to use.

Data Sourcing and Risk of Use

The data was sourced from open street map data, with some additional synthetic data being added to represent crime and lighting data. This creates the risk of inaccurate data being used to determine if a route is safe. To counteract this, the web app shows the disclaimers "This tool does not reflect real world data" and "This tool is not suitable for emergency use" to ensure that the user does not perceive the safety of the routes given to be absolute. We collected road data, as well as traffic, elevation, and distance between road nodes. Open street maps allow open use of data if proper credit is given.

Credit has been given in the README.md file in the program, and the acknowledgements section of this report.

Disclaimers

Route time predictions may vary due to unexpected real-world events that may be encountered along the route.

Remember to be fully aware of your surroundings at all times whilst using the app.

This version of the app only contains street data for Exeter UK, use outside of this area is not possible.

Routes are calculated based on indicator scores as well as distance, so the given route may not always be the most efficient route.

Crime / Lighting data used to calculate routes is synthetic and may not reflect reality.